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Artificial Intelligence & Data Science Department

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PaperGen

Synopsis by

(Group no. 05)

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I. TITLE

PaperGen

AI-Integrated Platform for Dynamic Learning Material Creation

PaperGen is an AI-powered tool designed to assist educators by automatically generating diverse academic content, such as quizzes and mini-projects, based on a provided syllabus or topic. The system uses a fine-tuned GPT-2 language model to create materials specifically tailored to different cognitive levels of Bloom's Taxonomy. By automating this traditionally time-consuming process, PaperGen aims to reduce teacher workload, enhance the quality and pedagogical alignment of educational resources, and ultimately facilitate a more efficient and personalized learning experience.

II. PROBLEM STATEMENT

The contemporary educational landscape places significant demands on educators, who are tasked with extensive curriculum development alongside their primary responsibilities of classroom instruction and assessment. A prevalent challenge is the time-consuming and often repetitive nature of designing high-quality academic materials, such as quizzes, presentations, and mini-projects. Specifically, teachers frequently encounter difficulties in crafting questions that precisely correspond to the various cognitive levels defined by Bloom’s Taxonomy (e.g., Remember, Apply, Create). Despite readily available learning objectives, the conversion of these objectives into Bloom-level-aligned questions necessitates considerable time, specialized training, and sustained effort.

III. Motivation

Recent advancements in the field of Natural Language Processing (NLP), especially with the emergence of sophisticated Large Language Models (LLMs), have fundamentally reshaped our understanding of what's possible in automated text generation. These powerful models can now understand context, nuances, and specific instructions to generate high-quality, human-like text at an unprecedented scale. This technological leap presents a significant opportunity to transform how academic content is created, moving from a labor-intensive, manual process to a scalable, on-demand system.

PaperGen is directly motivated by this potential, seeking to harness these technological strides to address the systemic challenges faced by educators. By leveraging intelligent automation, the project aims to do more than just create content; it seeks to fundamentally alleviate the immense workload on teachers, ensure a high level of quality and consistency in educational materials, and facilitate a more personalized approach to education delivery by providing tools to quickly generate tailored resources.

IV. Project Goal and Solution Overview

The primary goal of the PaperGen project is to develop an AI-based content generation tool that accepts a syllabus or topic as input and produces structured academic material tailored to different cognitive levels. This will be achieved by fine-tuning an open-source GPT-2 language model. PaperGen aims to serve as a comprehensive solution for educators, enabling them to efficiently generate diverse academic content that adheres to specific pedagogical requirements.

V. Proposed Methodology

The development of PaperGen will follow a structured approach encompassing several key phases:

- **Input Collection:** The system will allow users to upload or input syllabus content or specific topics. Users will then select the desired output type (Quiz Questions, Presentation Topics, or Mini Projects) and specify the target cognitive level from Bloom's Taxonomy (e.g., Remember, Apply, Create).

- **Model Development:** An open-source GPT-2 model will be fine-tuned using academic datasets that have been pre-tagged according to Bloom's Taxonomy levels. This fine-tuning process will utilize either PyTorch or TensorFlow. Furthermore, specific prompts and control templates will be designed to guide the model, ensuring the generation of domain-specific and level-appropriate academic content.

- **Frontend Implementation:** A user-friendly interface will be developed using React.js, incorporating form controls for syllabus uploads, output type selection, quantity specification, and Bloom's level selection. The frontend will also manage dynamic loading and provide a scrollable rendering of the generated output. TailwindCSS and Shadcn/UI will be used for styling.

- **Backend Implementation:** A RESTful API will be built using FastAPI (Python), leveraging its asynchronous capabilities. This backend will be responsible for receiving user input, invoking the fine-tuned GPT-2 model for inference, and returning the generated results. PostgreSQL will serve as the database for storing input-output data, generation logs, and potentially user profiles.

- **Evaluation & Output Formatting:** The generated content will be organized into logical categories (e.g., Multiple Choice Questions for "Remember" level, case studies for "Apply" level). The application will also provide export functionality, allowing teachers to download the generated materials in formats such as PDF or JSON.

VI. Expected Outcomes and Deliverables

Upon completion, PaperGen is expected to deliver a robust and fully functional web application that significantly aids educators in content creation. The core deliverables include:

- **A Fully Functional Web Application (PaperGen):** This will be the central product, providing an intuitive interface for users to interact with the AI content generation engine. It will be capable of:
    - **Accepting Academic Input and Parameters:** Users will be able to seamlessly upload or paste syllabus content, define specific topics, and set parameters such as the desired output type (quizzes, presentations, mini-projects) and the target Bloom's Taxonomy cognitive level.
    - **Generating Bloom-Level-Based Academic Content:** The application's primary function will be to produce high-quality, relevant academic materials that are precisely aligned with the chosen cognitive levels (e.g., generating recall-based questions for "Remember," problem-solving scenarios for "Apply," or creative tasks for "Create").
    - **Offering Downloadable and Structured Output:** The generated content will be presented in a clear, organized manner within the application and will be available for export in widely used formats like PDF for easy printing and distribution, and JSON for programmatic integration or further processing.

VII. Tools and Technologies

The development of PaperGen will leverage a carefully selected stack of modern and efficient tools and technologies, each chosen for its specific strengths:

A) Frontend:

- React.js: A popular JavaScript library for building user interfaces, known for its component-based architecture and efficiency in creating dynamic and interactive web applications.
- Tailwind CSS: A utility-first CSS framework that enables rapid UI development by providing low-level utility classes directly in the HTML, promoting consistent design and highly customizable styling.

B) Backend:

- Fast API (Python): A modern, fast (high-performance) web framework for building APIs with Python 3.7+ based on standard Python type hints. It's known for its excellent documentation, automatic data validation, and asynchronous support, making it ideal for high-throughput applications.
- PostgreSQL: A powerful, open-source object-relational database system known for its strong reliability, feature robustness, and performance. It will be used for persistent storage of user input and generated content.

C) Language Model:

- GPT-2 (Hugging Face Transformers): An open-source transformer-based language model developed by OpenAI, capable of generating coherent and diverse text. It will be fine-tuned using the Hugging Face Transformers.
- PyTorch or TensorFlow: Leading open-source machine learning frameworks that provide extensive libraries and tools for building, training, and deploying deep learning models. These will be used for the fine-tuning process.

D) Deployment:

- Render / Railway / Vercel / Hugging Face Spaces: Cloud platforms that simplify the deployment and hosting of web applications and machine learning models, offering features like continuous deployment, automatic scaling, and managed infrastructure.